RESEARCH STATEMENT

BALÁZS ELEK

1. My interests

I am interested in the representation theory of algebraic groups and the algebraic geometry surrounding them. My work often involves studying quiver varieties, flag manifolds and associated spaces.

I am especially interested in the geometry of Schubert varieties. Schubert varieties have connections to a wide range of other subjects, including representation theory and algebraic topology. I have several projects in this direction, including the ones I describe in section 2.1, section 2.2, section 3.2 and section 3.3.

Some of my work has a flavor of combinatorial algebraic geometry, and it involves turning interesting problems in geometry or algebra into combinatorics and studying them through some combinatorial gadget. Some historical examples where this approach has been successful are the convexity theorem for moment maps of Atiyah [Ati82], and independently, of Guillemin–Sternberg [GS82] and Kashiwara's crystal bases [Kas91], also independently discovered by Lusztig [Lus90] as the canonical bases. My projects described in section 3.1, section 3.3 and section 3.6 are examples of this.

I am also broadly interested in geometric representation theory. Geometric representation theory involves constructing geometric spaces and realizing various algebraic structures using them, often with the tools coming from homology theories. Some examples where this approach has been extremely successful are the proofs of the Kazhdan–Lusztig conjectures [KL79] by Brylinski–Kashiwara [BK80] and Beilinson–Bernstein [BiB81] via localization, or Nakajima's construction of the universal enveloping algebra using quiver varieties [Nak98]. The project in section 3.1 is an example of my research in this direction.

I will now list some of my current projects that are in an advanced stage below.

2. Current projects

2.1. **Algebraic Double loop spaces of Flag varieties.** In a project joint with Jim Bryan, Freddie Manners and Ravi Vakil, I am studying the geometry and topology of the space

$$\Omega^2_\beta(\mathrm{Fl}_{n+1}) = \{f : \mathbb{P}^1 \to \mathrm{Fl}_{n+1} \mid f_*([\mathbb{P}^1]) = \beta, f([1:0]) = [\mathrm{Id}]\}$$

of genus zero based maps in the class $\beta \in H^2(\mathrm{Fl}_{n+1})$ into a full flag variety.

Our motivation for studying this space come from Bott periodicity [Bot59].

By a Theorem of Boyer–Mann–Hurtubise–Milgram [BMHM94], as the class β tends to infinity in a suitable sense, the space $\Omega^2_\beta(\mathrm{Fl}_{n+1})$ can be viewed as a homotopy approximation to the *topological* double loop space $\Omega^2_{\beta,\mathrm{top}}(\mathrm{Fl}_{n+1})$. There space $\Omega^2_{\beta,\mathrm{top}}(\mathrm{Fl}_{n+1})$ is homotopy equivalent to $\Omega^2_{\mathrm{top}}(U(n+1))$, and the *rational* homotopy type of $\Omega^2_{\mathrm{top}}(U(n+1))$ is known

$$\Omega^2_{\beta,\mathrm{top}}(U(n+1)) \simeq_{\mathbb{Q}} U(n).$$

Question 2.1. How does the rational homotopy type of $\Omega^2_\beta(\mathrm{Fl}_{n+1})$ compare to the rational homotopy type of $U(n)$?

The space $\Omega^2_\beta(\mathrm{Fl}_{n+1})$ is an open subset inside several well-studied spaces in geometric representation theory, for example the Drinfeld Zastava spaces studied by Finkelberg and Rybnikov [FR14] and Handsaw quiver varieties studied by Nakajima [Nak12]. It would be interesting to explore the implications of our results in this direction.

Expand $\beta = \sum_{i=1}^n d_i[X_{s_i}]$ in the Schubert basis. We call the class β **monotonic** if $d_i > d_{i+1}$. For β monotonic, Jim Bryan and I discovered a remarkable canonical basis for the space $H^0(\mathcal{E}_1(-1))$, where $\mathcal{E}_1$ is the pullback of the universal line bundle on Fl_{n+1}. This basis leads to a completely explicit description of $\Omega^2_\beta(\mathrm{Fl}_{n+1})$ as an affine variety (in fact, a hypersurface in affine space). The defining equation is quite complicated, but it still led us to the following:

Theorem 2.2 ([BE25])**.** *In the minimial monotonic case ($\beta = \sum_{i=1}^n (n-i+1)[X_{s_i}]$) there is an isomorphism of varieties*

$$\Omega^2_\beta(\mathrm{Fl}_{n+1}) \cong \mathrm{GL}(n) \times \mathbb{C}^n.$$

Following a suggestion by Ravi Vakil, using `Macaulay2` [GS], I computed some point counts of $\Omega^2_\beta(\mathrm{Fl}_{n+1})$ for monotonic β over finite fields, and these agreed with the point counts of $\mathrm{GL}(n) \times \mathbb{A}^D$ (for some D depending on β). We were then able to prove the following:

Theorem 2.3 ([BEM+26])**.** *Suppose $\beta = \sum_{i=1}^n d_i[X_{s_i}]$ is monotonic. Then we have the following equality in $K_0(\mathrm{Var})$, the Grothendieck group of varieties over :*

$$\left[\Omega^2_\beta(\mathrm{Fl}_{n+1})\right] = \left[\mathrm{GL}(n) \times \mathbb{A}^{D-n^2}\right]$$

where $D = \sum_{k=1}^n 2d_k$.

Based on the above results, we make the following conjecture:

Conjecture 2.4. *For monotonic classes β, the inclusion map $\Omega^2_\beta(\mathrm{Fl}_{n+1}) \hookrightarrow \Omega^2_{\beta,\top}(\mathrm{Fl}_{n+1})$ induces an isomorphism of rings:*

$$H^*(\Omega^2_\beta(\mathrm{Fl}_{n+1}), \mathbb{Q}) \cong H^*(U(n), \mathbb{Q}).$$

This project also involved interesting interactions with Google Deepmind and generative AI. Deepmind wanted to test the capability of some of their internal models on research problems and we were curious and agreed that this project could be a test case. With guidance and scaffolding from us, the model was able to produce a proof of the point count statement that included some ideas that we were then able to use to prove theorem 2.3.

2.2. **Hypercube Richardson varieties.** Let $W = S_n$ be a type A Weyl group. In a recent preprint [ELP+26], Ellenberg, Libedinsky, Plaza Simental and Williamson describe Bruhat intervals in W that are large hypercubes (boolean posets). The authors contextualize this result in the framework of trying to prove the combinatorial invariance conjecture of Dyer and Lusztig that says that the Kazhdan–Lusztig polynomials are determined by the combinatorial structure of the interval as a poset. In particular, it is known that combinatorial invariance holds for intervals that are hypercubes.

The geometric object associated to a Bruhat interval $[v, w]$ is the Richardson variety $X_v^w = X_v \cap X^w$ where $X_v = \overline{B^- vB/B}$ is an opposite Schubert variety and $X^w = \overline{BwB/B}$ is a Schubert variety. I am working on precisely understanding the geometry of Richardson varieties where the Bruhat interval is a hypercube. In particular, I conjecture that such a Richardson variety is always isomorphic to a tower of $\mathbb{P}^1$-s.

Let Q be a reduced word for w. Say that Q is a **magic word** for $[v, w]$ if Q has a unique subword with product equal to v. It is not hard to see that if Q is a magic word for $[v, w]$ then the interval must be a hypercube, but the converse is not obvious.

Conjecture 2.5. *Every interval $[v, w]$ that is a hypercube has a magic word.*

I have verified this conjecture for all intervals up and including type A_6, and I have some ideas on how to prove it.

Let $Q = (s_1, \ldots, s_k)$ be a word in W. Let $P_{s_i} = B \cup Bs_iB$. The **Bott-Samelson variety** BS^Q is the quotient of the product manifold $P_{s_1} \times \ldots \times P_{s_k}$ by the right action of B^n by

$$(p_1, p_2, \ldots, p_n) \cdot (b_1, b_2, \ldots, b_n) = (p_1 b_1, b_1^{-1} p_2 b_2, \ldots, b_{n-1}^{-1} p_n b_n).$$

Then BS^Q is an iterated $\mathbb{P}^1$-bundle with a canonical map m to G/B, given by multiplying the p_i-s. Bott-Samelson varieties have been used as effective tools in answering important questions in representation theory and the geometry of Schubert varieties, see, for example [Bri05] and [EW14].

It is not difficult to show that if Q is a magic word for $[v, w]$, then the natural map $m : BS^Q \to G/B$ is one-to-one when restricted to $m^{-1}(X_v)$. Since the image $m(BS^Q)$ is the Schubert variety X^w, the Richardson variety X_v^w is isomorphic to $m^{-1}(X_v^w)$. I have proved the following theorem, which gives a complete description of the Richardson variety X_v^w (an early preprint is available upon request).

Theorem 2.6. *Let Q be a magic word for $[v, w]$. Then the preimage $m^{-1}(X_v^w)$ inside BS^Q is given explicitly by setting $p_i = s_i$ for those i that are included in the unique subword with product v.*

It is worth pointing out that conjecture 2.5 would supply a new (geometric) proof of the combinatorial invariance conjecture for the case of Bruhat intervals that are hypercubes (the conjecture is known to be true in this case), since an iterated $\mathbb{P}^1$-bundle is smooth, for smooth Richardson variety, the Kazhdan–Lusztig polynomials are constant 1.

2.3. **(Hyper-)Brick manifolds and Braid varieties.** This is a joint project with Raj Gandhi. Let G be a semisimple algebraic group with a fixed Borel subgroup B. Let $Q = (s_1, \dots, s_n)$ be a (not necessarily reduced) word in W, the Weyl group. Then we have the Bott–Samelson variety BS^Q as defined in section 2.2.

The map m is surjective if there is a subword of Q whose product is the long element $w_0 \in W$. We will assume that this is the case in this section. The **brick manifold**, introduced and studied by Escobar in [Esc16], is the fiber $\mathrm{Brick}(Q) = m^{-1}(w_0 B/B)$. Let N denote the unipotent radical of B. Since the map m is B-equivariant, we have an N-equivariant isomorphism

$$N \times \mathrm{Brick}(Q) \cong m^{-1}(N w_0 B/B).$$

Note that since $N w_0 B/B$ is open in the flag variety, $m^{-1}(N w_0 B/B)$ is open in BS^Q and carries a free action of N, so this suggests that one could define

$$BS^Q /\!/ N := m^{-1}(N w_0 B/B)/N = \mathrm{Brick}(Q)$$

and think of the brick manifold as a GIT quotient of the Bott-Samelson variety. This motivated Allen Knutson to ask the question

Question 2.7. What is the the HyperKähler analogue of the brick manifold? In other words, can we find a reasonable definition for

$$T^*(BS^Q)////N?$$

Non-reductive GIT quotients are notoriously difficult to deal with [DK07]. In particular, rings of invariants are not guaranteed to be finitely generated [Nag59]. In fact, in our situation, even for $G = SL(2)$, the ring $\mathbb{C}[\mu^{-1}(0)]^N$ (where $\mu : T^*(BS^Q) \to \mathfrak{n}^*$ is the moment map) is not finitely generated.

One way of circumventing this non-Noetherian problem is to use an N-action that extends to a G-action where $N \subseteq G$ and G is reductive. We consider the invariant ring $\mathbb{C}[T^*(BS^Q)]^N$ and impose the moment map condition *afterwards*. We have some promising preliminary results in this direction for $G = SL(2)$.

During the course of the project we used `Macaulay2` [GS] extensively to compute examples. In the case of $G = SL(2)$, our preliminary definition for quotients $BS^Q /\!/_\chi N$ depends on the choice of a character χ of a torus $(\mathbb{C}^\times)^n$, and we know that for a certain choice of χ, the quotient is the brick manifold, but for other choices, we obtain other projective varieties.

Brick manifolds are also natural compactifications of **braid varieties**, which are of independent interest [CGG+25][CGGS24]. Ian Le suggested that maybe these other projective

varieties we obtained could be compactifications of braid varieties, and we have recently been able to confirm this in the $SL(2)$ case.

3. Previous work

3.1. **Quiver variety components, heaps and minuscule combinatorics.** In [KS97], Kashiwara and Saito defined a crystal structure on the set of irreducible components of Lusztig quiver varieties. In [Sav06], Savage showed that in types A and D this coincides with the combinatorial definition of crystals, establishing a link between the geometry of quiver varieties and the combinatorics of crystals.

In a similar flavor, joint with Anne Dranowski, Joel Kamnitzer, Tanny Libman and Calder Morton-Ferguson, I connected certain modules of the preprojective algebra of an ADE type Dynkin quiver to crystals [DEKM22].

Let W denote the Weyl group and let P be a maximal parabolic subgroup corresponding to a minuscule fundamental weight ω. Let $w = w_0^P$ denote the minimal length representative of $w_0 W_P$ where W_P is P's Weyl group. Our construction works in a slightly more general setting, the technical assumption we need on w is **dominant minuscule** [Ste01], but we'll focus on the w_0^P case for simplicity.

Consider the **heap** $H(w)$ ([Ste96]) of w. This is a partially ordered set with a map to the Dynkin quiver. Let k be a positive integer. Using $H(w)$ one can construct a combinatorial model $RPP(w, k)$ for the crystal $B(k\omega)$ (elements of this model are called **reverse plane partitions** and have been defined before, see [GPT18], for example).

We also construct space of modules $L(k\omega)$ for the preprojective algebra of the Dynkin quiver. We use Nakajima's tensor product varieties ([Nak01]) to establish an explicit crystal isomorphism between $RPP(w, k)$ and the irreducible components of the preprojective algebra modules.

We also made some conjectures regarding the action of cactus and toggle groups (see Section 2.8 of [DEKM22]. Recently, joint with Devin Brown and Iva Halacheva in [BEH24], I was able to prove some of these conjectures, which extend the results of Chmutov, Glick and Pylyavskyy [CGP16] on Berenstein-Kirillov groups and cactus groups.

3.2. **Affine Kazhdan–Lusztig Varieties.** Let v and w be permutations in S_n, and let $X_\circ^w = BwB/B$ be a Bruhat cell and $X_v = \overline{B^- vB/B}$ an opposite Schubert variety. In [KL79], Kazhdan and Lusztig introduced **Kazhdan–Lusztig varieties** $\mathcal{X}_{v,\circ}^w = X_\circ^w \cap X_v$. They have since found numerous applications, for example, they can be used to study the geometry of Schubert varieties in the neighborhood of a torus-fixed point. Since the Bruhat cell $X_\circ^w$ is just an affine space $\mathbb{C}^{l(w)}$, it is natural to study the **Kazhdan–Lusztig ideal**, the ideal defining $X_{v,\circ}^w$ inside $X_\circ^w$. In [WY12], Woo and Yong gave a Gröbner basis for Kazhdan–Lusztig ideals for the type A flag variety. Having a Gröbner basis of any interesting ideal is desirable because it can simplify computation, but even better, for Kazhdan–Lusztig ideals, the Gröbner basis has square-free leading terms, and therefore can be used to

degenerate the Kazhdan–Lusztig variety into a reduced union of coordinate subspaces, and this union is described by a **subword complex** of Knutson and Miller [KM05].

Joint with Daoji Huang, I extended Woo and Yong's results to affine type A. Building on previous work of Huang [Hua19], we present a Gröbner basis for Kazhdan–Lusztig ideals. We develop some tools that are well known in the finite type A case, including a generalization of Fulton's essential set [Ful92] and a linear parametrization for the Bruhat cell $X_\circ^w$. We then adapt the proof in [WY12] to the affine case.

We hope that our results can be used to practically compute with equations defining Kazhdan–Lusztig varieties in affine type A using software, and we have used `Macaulay2` [GS] extensively. We also hope that our result can be useful in studying singularities of Schubert varieties in affine type A.

3.3. **My thesis work.** Motivated by the combinatorics of the positroid stratifications of Grassmannians in [KLS13] and the corresponding geometry in [Sni10], He, Knutson and Lu introduced Bruhat atlases in [HKL]. A **Bruhat atlas** on a stratified manifold $(M, \mathcal{Y})$ is a way to model the stratification $\mathcal{Y}$ on the Bruhat stratification. Specifically it is an open cover of M by Schubert cells X_0^w in a flag manifold H/B_H (of a Kac-Moody group H) compatible with the stratification $\mathcal{Y}$ and with the opposite Schubert stratifications of the X_0^w's. In [KLS13], Bruhat atlases are described on the wonderful compactifications $\overline{G}$ of DeConcini-Procesi [DCP83], and on partial flag manifolds G/P of semisimple algebraic groups. The existence of Bruhat atlases on these two interesting families of varieties leads one to ask the question: Can one classify manifolds with Bruhat atlases?

I worked on the classification of toric surfaces with Bruhat atlases compatible with the torus action, with complete results in the simply-laced case. This project is in the intersection of combinatorics, representation theory and algebraic geometry, and the proofs involved a broad range of methods. In [Ele16], I proved that the only smooth toric surfaces with Bruhat atlases are $\mathbb{CP}^1 \times \mathbb{CP}^1$ and $\mathbb{CP}^2$, but there are more varieties with a so-called **Kazhdan–Lusztig atlas**, defined also in [KLS13]. I classified smooth toric surfaces with simply-laced Kazhdan–Lusztig atlases in my thesis ([Ele16]). This involved a number of steps, with surprisingly different techniques for each one.

- Classifying the moment polygons of smooth Richardson surfaces in arbitrary Kac-Moody groups. There are 10 of these, plus an infinite family.
- Assembling these into a polygon of the toric surface. I reduced this to a finite problem by considering the abelianization map from $SL(2, \mathbb{Z})$ to $\mathbb{Z}/12\mathbb{Z}$, then used Sage ([Dev15]) to obtain the 20 possible configurations, which I called pizzas.
- Finding a Kac-Moody group H such that a given pizza appears in H/B_H. I found these by building a root system from necessary conditions for the equivariant cohomology classes of the 1-skeleton of the pizza, which came from the structure of Bruhat intervals of height 2 and 3.

Using the work of Dyer [Dye91], I also proved that there are at most 7543 non-simply laced pizzas with Kazhdan–Lusztig atlases, each coming from a subdivision of a polygon with at most 12 vertices.

3.4. **Finite type multiple flag varieties.** In [MWZ00] and [MWZ99] Magyar, Weyman and Zelevinsky consider G-actions on multiple flag varieties $G/P_1 \times \ldots \times G/P_k$ (for $P_1, \ldots, P_k$ parabolic subgroups of $GL(n)$ or $Sp(n)$) and answer the question of when such a G-action has finitely many orbits. Several special cases of this have been considered, for instance, the spherical case (i.e. when $P_1 = B$) in [AP13], [Lit94], [Ste03], and the case $P = P_1 = P_2 = P_3$ in [Dev14], [Pop07].

Joint with with Dan Barbasch, Sergio Da Silva and Gautam Gopal Krishnan, I extended the above results to exceptional groups. In [MWZ00], the authors exploit the description of GL_n/P as the variety of partial flags in $\mathbb{C}^n$ to transform the question to quiver representations, and in [MWZ99], they reduce the type C cases to the type A case. This approach does not obviously generalize to exceptional groups, so we took a more direct approach. The G_2 case is trivial, as by a crude dimension count one notices that G can have finitely many orbits only when $k = 2$, in which case the question is completely answered by the Bruhat stratification on $G/P_2 = \bigsqcup_{w \in P_1 \backslash W / P_2} P_1 w P_2 / P_2$. We finished the F_4 case in [BDSEK17] and are currently working on the type E cases.

3.5. **The standard Poisson structure on Bott-Samelson varieties.** Let G be a semisimple Lie Group over $\mathbb{C}$ and let $Q = (s_1, \ldots, s_k)$ be a word in the simple reflections of W. Let BS^Q be the Bott–Samelson variety as defined in section 2.2.

A choice of a pair $T \subseteq B$ of a maximal torus T, a Borel subgroup B and a symmetric non-degenerate biliear form gives rise to a multiplicative holomorphic Poisson structure π_{st} on G, making (G, π_{st}) into a Poisson-Lie group that is the semi-classical limit of the quantum group associated to G [CP94]. Since B is a Poisson Lie subgroup of G, this projects to a well-defined Poisson structure $\pi_{G/B}$ on G/B. Similarly, since all the P_{s_i}'s are Poisson subgroups, there is a standard Poisson structure π_Q on BS^Q and the map $m : BS^Q \to G/B$ is Poisson.

Since BS^Q is an iterated $\mathbb{P}^1$-bundle, it has many natural coordinate charts. Joint with Jiang-Hua Lu, in [EL19] I explicitly computed the standard poisson structure in each of these charts. The formulas are given in terms of root strings in the root system of G, and are entirely combinatorial, and I implemented their computation in GAP [GAP19].

We showed that he Poisson structure in each of the charts is a Poisson polynomial algebra, and that in one of the charts, it is a **symmetric Poisson CGL extension**, studied by Goodearl and Yakimov [GY14]. Symmetric Poisson CGL extensions have deep connections to the theory of cluster algebras, see, for example [GSV10]. In particular, in [GY16], Goodearl and Yakimov prove the Berenstein-Zelevinsky conjectures on the equality of the cluster algebras and upper cluster algebras associated to double Bruhat cells using Poisson CGL extensions.

Our results in [EL19] also have applications to quantum groups, integrable systems, total positivity and toric degenerations of some Poisson varieties associated to G. See also the introduction of [EL19] for some directions for future research.

3.6. **Promotion and cyclic sieving on δ-semistandard tableaux.** Let X be a finite set with an action of the cyclic group $C_l = \langle c \rangle$, ζ a primitive lth root of unity and $f(q)$ a polynomial. Then the triple $(X, \langle c \rangle, f(q)$ **exhibits the cyclic sieving phenomenon (CSP)** (introduced by Reiner, Stanton and White in [RSW04]) if $f(\zeta^d) = |X^{c^d}|$ for all integers $d \geq 0$ where X^{c^d} denotes the fixed-point set of c^d. One can show that such a polynomial always exists, but the interest in CSP is that often the polynomial $f(q)$ is a q-deformation of a formula for enumerating $|X|$. For example, in [RSW04], Reiner, Stanton and White show that if X is the set of all subsets of $\{1, 2, \ldots, n\}$ of size k, and C_n acts on X via the long cycle $(1, 2, \ldots, n) \in S_n$, then $\left(X, C_n, \begin{bmatrix} n \\ k \end{bmatrix}_q\right)$ exhibits the CSP, where $\begin{bmatrix} n \\ k \end{bmatrix}_q$ is the q-analogue of the binomial coefficient. Note that in this case $|X| = \binom{n}{k}$, the ordinary binomial coefficient.

In [Rho10], Rhoades proves several CSP results about rectangular Young tableaux. It is well-known that on the set of Young tableaux with fixed rectangular shape with entries up to n, promotion has order n. The cyclic sieving polynomials are again q-deformations of natural counting formulae. In particular, Theorem 1.5. in [Rho10] establishes a cyclic sieving result on rectangular column-strict tableaux on fixed shape and content, where the cyclic sieving polynomial is the Kostka-Foulkes polynomial. In [FK14], Fontaine and Kamnitzer interpreted the vector space $\mathbb{C}X$ spanned by these tableaux as the space of invariants in a tensor product of minuscule representations of SL_n, where the promotion on tableaux corresponds to cyclic rotation of the tensor factors. Using the geometric Satake correspondence and Nakajima quiver varieties, they give a geometric proof and a generalization of Rhoades' cyclic sieving result.

Joint with Tair Akhmejanov, in [AE20], I extended Fontaine and Kamnitzer's results to a case where the geometric action is on an invariant space in a tensor product

$$V(\vec{\lambda}) = V(\lambda_1) \otimes V(\lambda_2) \otimes \ldots \otimes V(\lambda_n),$$

where each $V(\lambda_i)$ is either a symmetric or alternating power of the defining representation $\mathbb{C}^n$ (in this context, the result of Fontain and Kamnitzer is where all the representations are $\wedge^k \mathbb{C}^n$s).

We define a new class of tableaux, called δ**-semistandard tableaux**, to naturally index the basis for the invariant space $V(\vec{\lambda})^G$, where $\delta = (\delta_1, \ldots, \delta_n)$ is a binary string with δ_i specifying whether the entries equal to i of the tableau appear as a horizontal or vertical strip. Using the methods developed in Akhmejanov's thesis [Akh18], we are able to prove that the δ-analogue of the promotion operator still has order n, and we adapt the techniques of Fontaine and Kamnitzer to establish a CSP with the polynomial equal to the **generalized Kostka polynomials** introduced in [SW99] and [KS02].

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